

About:
– 2–cell to 7–cell battery managment system designed to handle and protect Li–Ion and Li–Po batteries.

Features:
– Estimates battery state of charge (SoC), state of health (SoH) and others
– Reads battery voltage, cells voltages and battery current
– Reads battery temperature (NTC) and ambient (board) temperature
– Balances battery cells
– Provides I2C and CAN communications to the user
– Powerdown mode consumes < 2uA

Protections – there are 3 levels of protection
– Software protections, MCU control, safely shutting down battery charge/discharge
– Hardware protection (TPS1200 switch) which detect battery overvoltage and overcurrent and shuts it down, independetly of MCU
– Fuse and TVS protection – in case that software and TPS1200 switch fails, there is a fuse which burns and physically disconnects the battery.

Revision B Change Log

Power Control & Protection

Add manual enable switch to bring the board out of shutdown, rather than relying solely on an external signal holding EN/UVLO high
Retune EN/UVLO and OV divider resistors to land both trip points at the intended pack voltages, and give the PMOS load–switch gate (Q3) adequate V_{GS} margin across the full input range
Review GND_BATT connection for voltage measuring paths

Analog Front–End & Measurement Accuracy

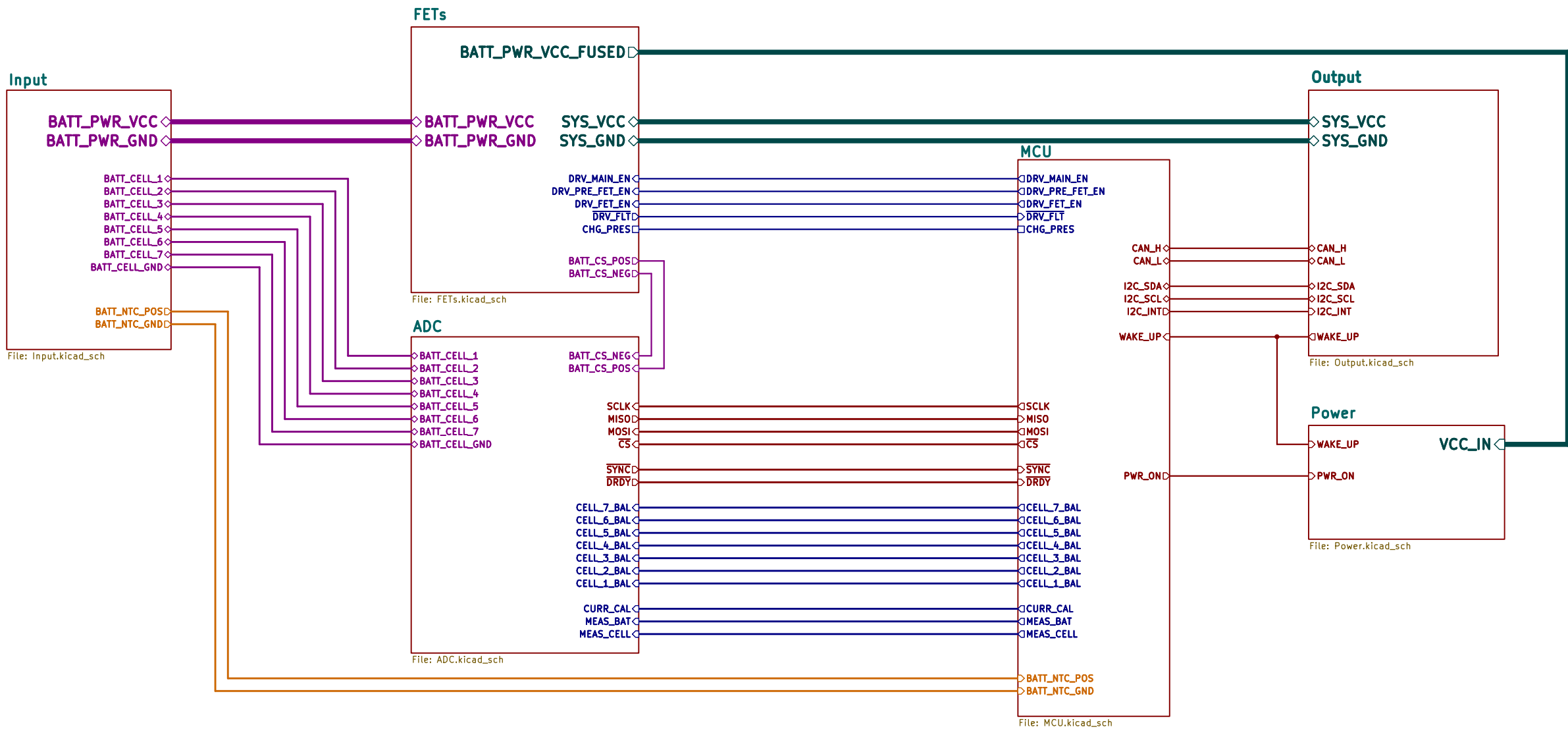
NTC divider polarity: fixed resistor was wired to GND instead of VCC, invalidating the temperature reading
Use 0.1% resistors on measurement dividers to reduce channel–to–channel gain mismatch and tighten protection–threshold accuracy
Measure SYS_VCC to detect charger presence directly, instead of inferring it from current direction
Add current–sense offset calibration path: switch to short the current–sense differential inputs so offset can be characterized independent of load current
Reference all measurement dividers to BATT_GND to remove ground–offset error between channels

Component & Schematic Corrections

LED package change: side–view LED replaced with a top–view package
Dual NMOS + PMOS schematic symbol didn’t match the physical part’s pinout – corrected
FET SRC–to–driver connection: driver’s source reference now matches the FET’s actual source node
EEPROM memory updated

PCB Layout

FET footprint–to–zone connection: power FET pads now properly stitched to copper pour for current–carrying capacity and thermal dissipation



Open Source Battery Management System
Designed by OpenBatt team
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Sheet: /
File: openbms.kicad_sch

Title: OpenBMS

Size: A3 Date: 2026–09–24
KiCad E.D.A. 9.0.7

Rev: B
Id: 1/7

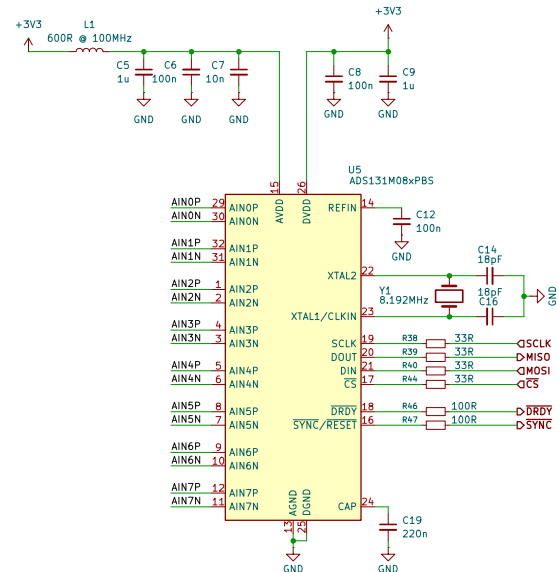
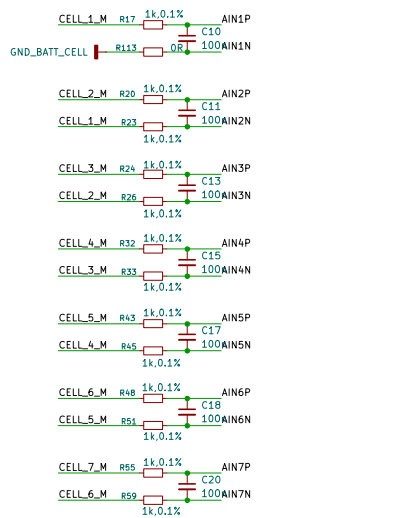
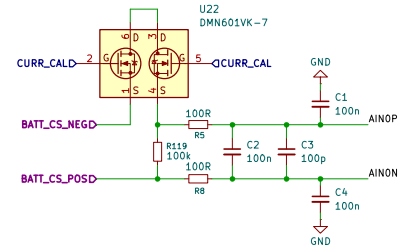
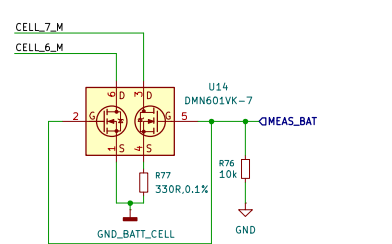
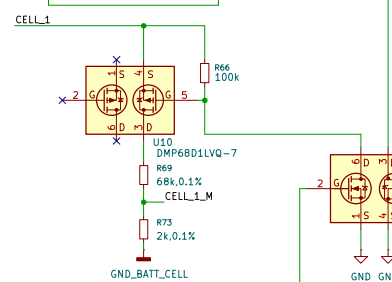
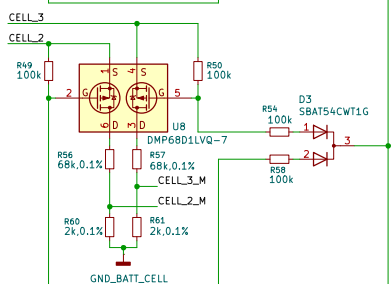
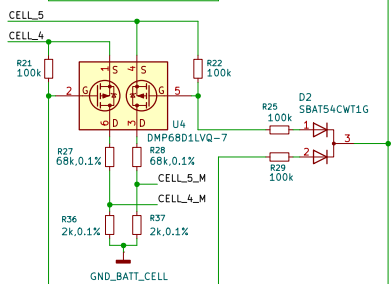
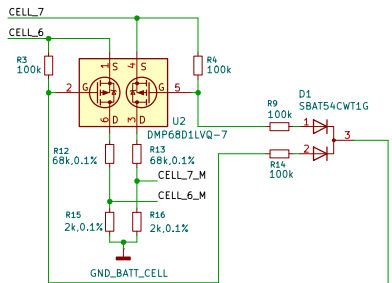
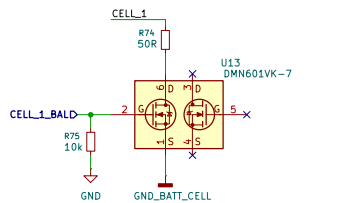
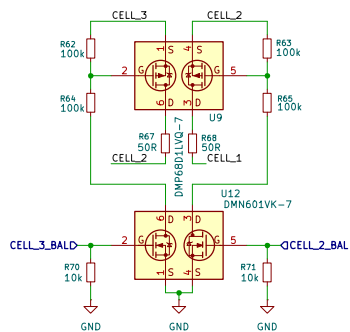
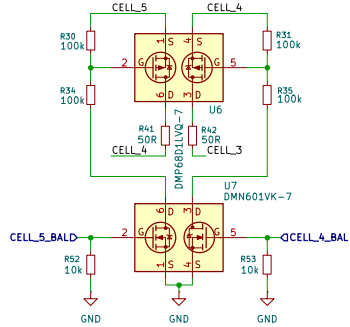
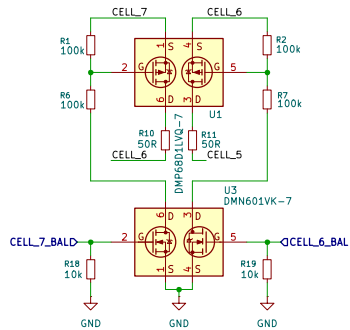
About:
~ ADC circuit for battery voltage and current reading
~ Cell FETs for cell balancing

Design Parameters:
~ 7-cell battery voltage monitoring (between 2.5V and 4.3V)
~ 7-cell battery balancing (up to 86mA per cell)
~ Current monitoring (up to 20A)

How to use it:
1) Battery cells are disabled from ADC by default to disable current leaking from battery when BMS is off. To enable battery reading, set MEAS_CELL to 1
2) Read differential AINxP - AINxN voltage and multiply with $(68+2)/2 = 35$ to get batter cell voltage in V
3) Read battery current by reading voltage between AIN0P and AIN0N. Divide voltage by 0.002 (shunt resisot is 2m ohm) to get current in A
4) There is a special case to read overall battery voltage by setting MEAS_BATT to 1 (while keeping MEAS_CELL at 1) and read voltage between AIN7P and AIN7N (like you would read cell 7)
5) Balancing circuit for each sell can be enabled individually by setting CELL_x_BAL to 1. While balancer (cell FET) is on, don't read battery voltage, as you could get less precise cell voltage

BATT_CELL_7<---CELL_7
BATT_CELL_6<---CELL_6
BATT_CELL_5<---CELL_5
BATT_CELL_4<---CELL_4
BATT_CELL_3<---CELL_3
BATT_CELL_2<---CELL_2
BATT_CELL_1<---CELL_1

BATT_CELL_GND<---GND_BATT_CELL



SCLK, MOSI and CS resistors place close to MCU.
other resistors place close to ADC1

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Sheet: /ADC/
File: ADC.kicad_sch

Title: OpenBMS

Size: A2 Date: 2026-09-24

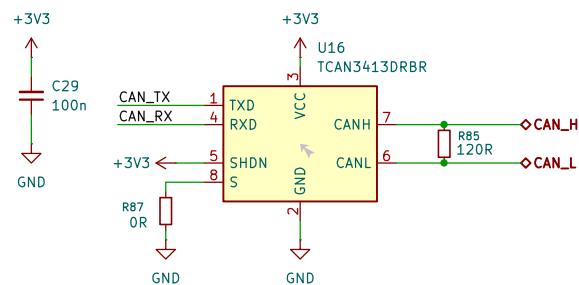
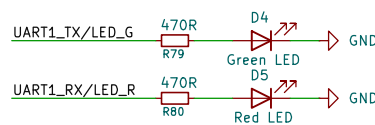
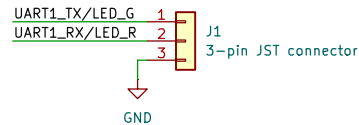
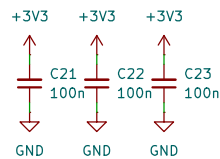
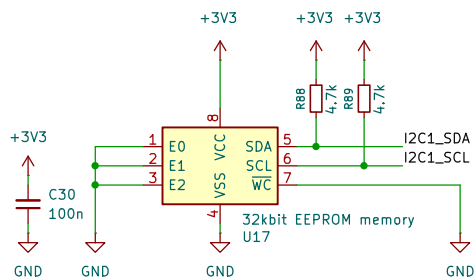
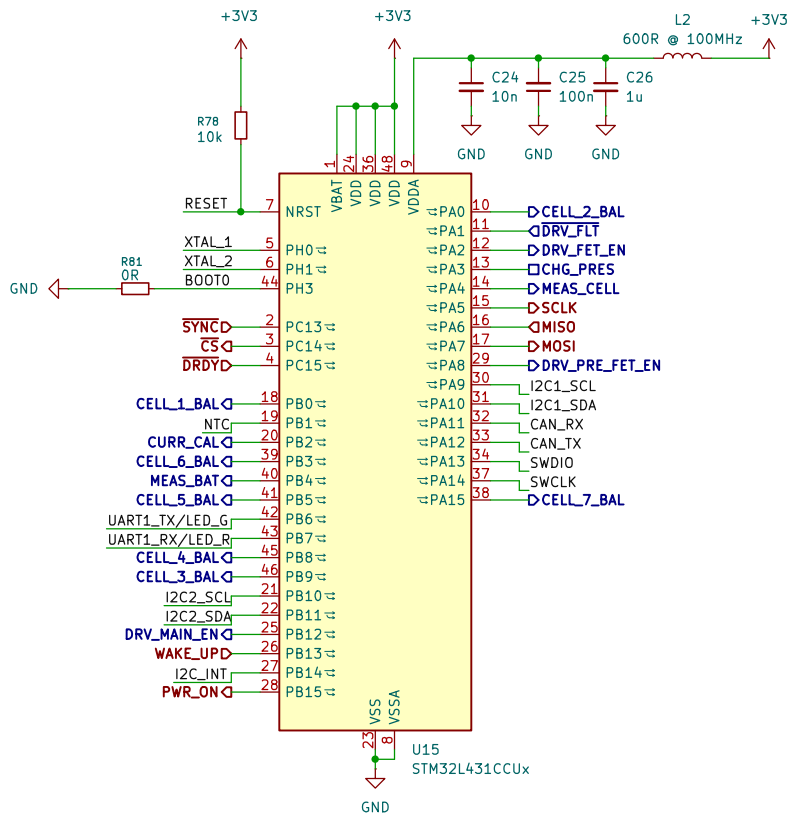
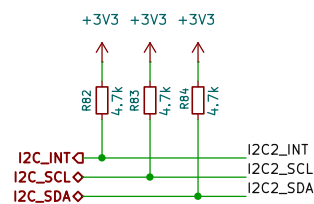
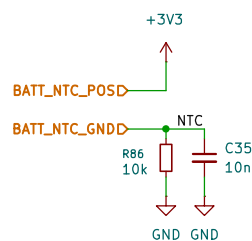
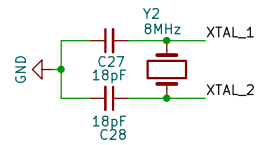
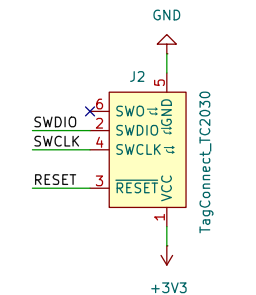
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Rev: B

Id: 2/7

About:
- STM32L43 MCU is system brain

Functionalities:
- Configure and reads data from ADC (ADS131M08)
- Controls balancer FETs
- Controls main switch (TPS1200)
- Reads battery NTC
- Stores battery data into EEPROM
- Handles user communication I2C and CAN
- Provides UART debug communication



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Sheet: /MCU/
File: MCU.kicad_sch

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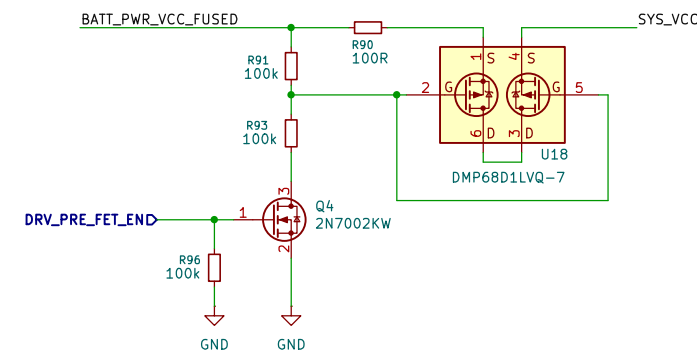
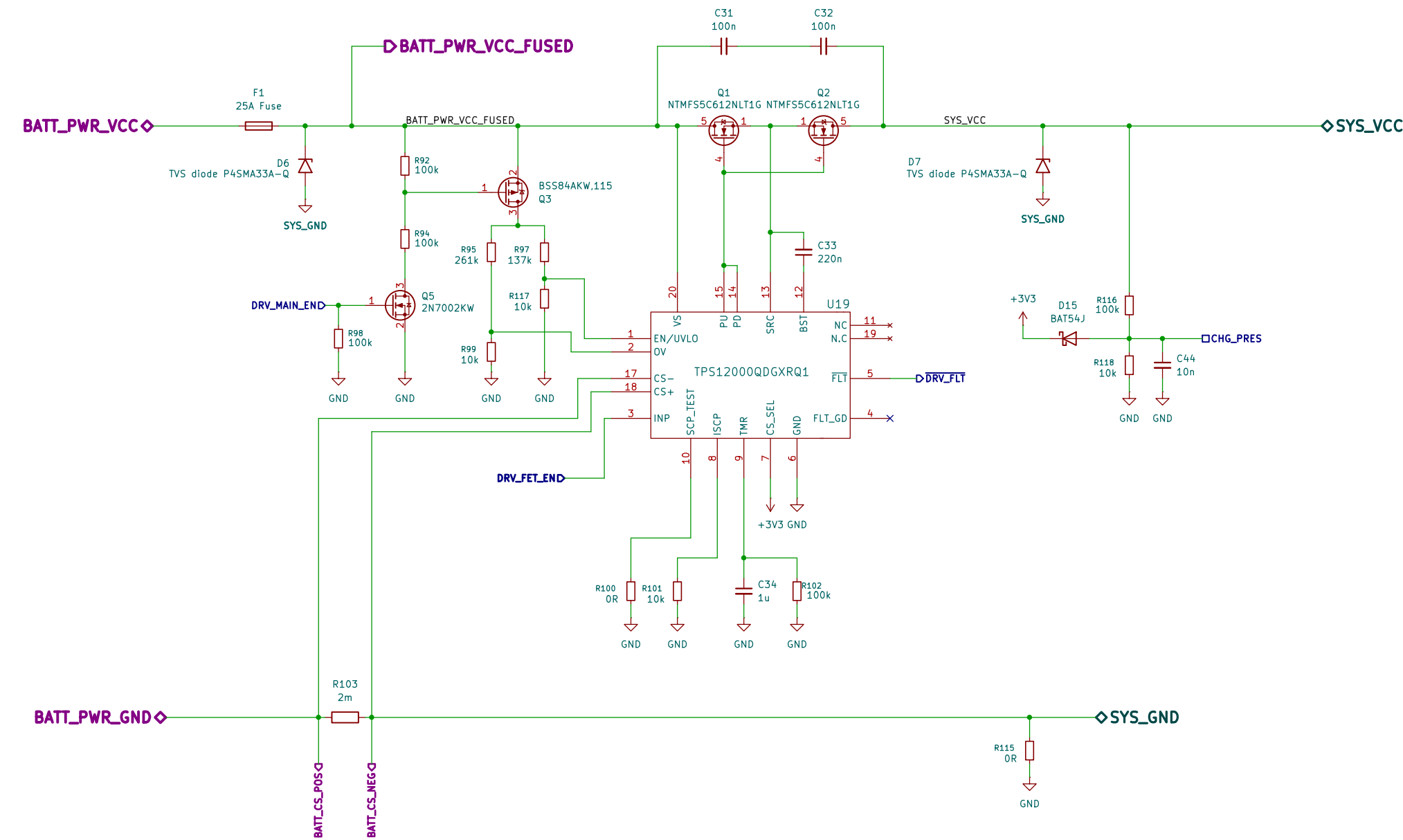
Size: A3 Date: 2026-09-24
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Rev: B
Id: 3/7

About:
- TPS1200 is main battery switch. It controller power FETs and protect against battery overvoltage, undervoltage and overcurrent

Design Parameters:
- 7-cell battery, minimum cut-off voltage $2.5V \times 7 = 17.5V$ (set by EN/UVLO and OV resistors)
- 7 cell battery, maximum cut-off voltage $4.3V \times 7 = 30.1V$ (set by EN/UVLO and OV resistors)
- Maximum battery cut-off current = 20A (set by shunt resistor and ISCP resistor)
- Burn-out fuse will blow at 25A within 0.3-0.6S

How to use it:
1) Set DRV_MAIN_EN to 1 to enable TPS1200
2) Set DRV_PRE_FET_EN to 1 to start precharge/predischarge (optional)
3) Set DRV_FET_EN to 1 to enable main FETs. This enables battery charge/discharge
4) Watch for DRV_FLT_GD. When 1, FET gate voltage is good
5) Watch for FLT. When 0, there is a fault present (overvoltage, undervoltage, excessive current..)



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Sheet: /FETs/
File: FETs.kicad_sch

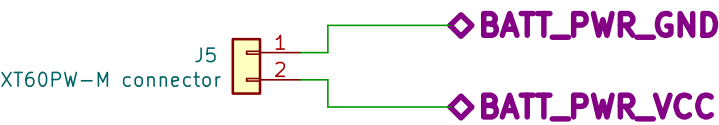
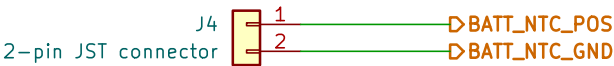
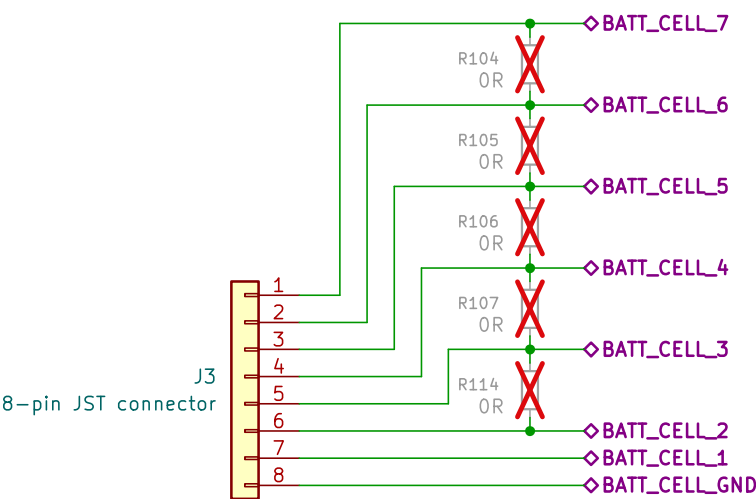
Title: OpenBMS

Size: A3 Date: 2026-09-24
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Rev: B
Id: 4/7

About:
– Input of the battery. XT60 power connector + 8-pin JST connector for cell wirse + 2-pin JST connector for battery NTC

Design info:
–If battery has less then 7 cells, then place jumper between cell connections which are not being used, e.g. for 5-cell battery jumpers between cells 6 and 7 and betwenn 5 and 6 need to be populated



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Sheet: /Input/
File: Input.kicad_sch

Title: OpenBMS

Size: A4

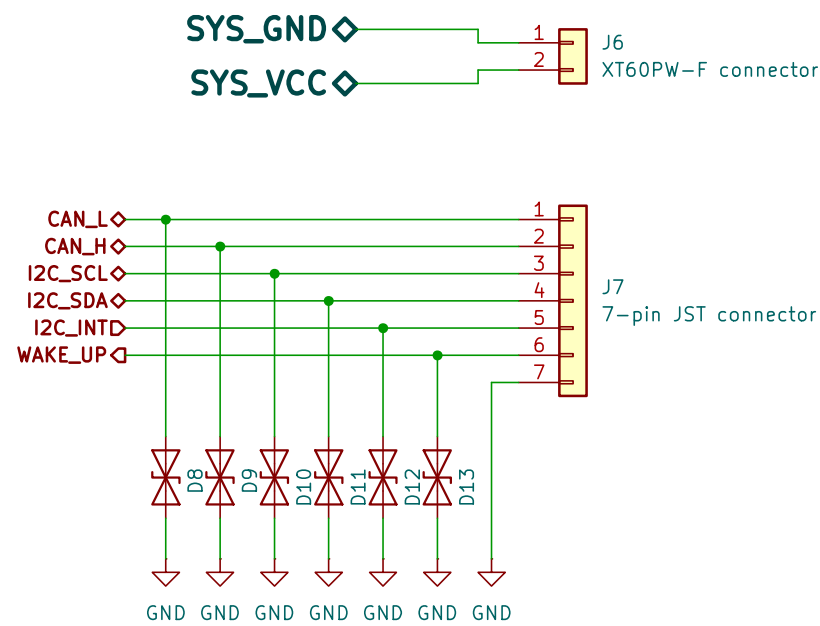
Date: 2026-09-24

Rev: B

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Id: 5/7

About:
– Output of the BMS, power XT60 connector + JST connector with signals



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Sheet: /Output/
File: Output.kicad_sch

Title: OpenBMS

Size: A4 Date: 2026-09-24

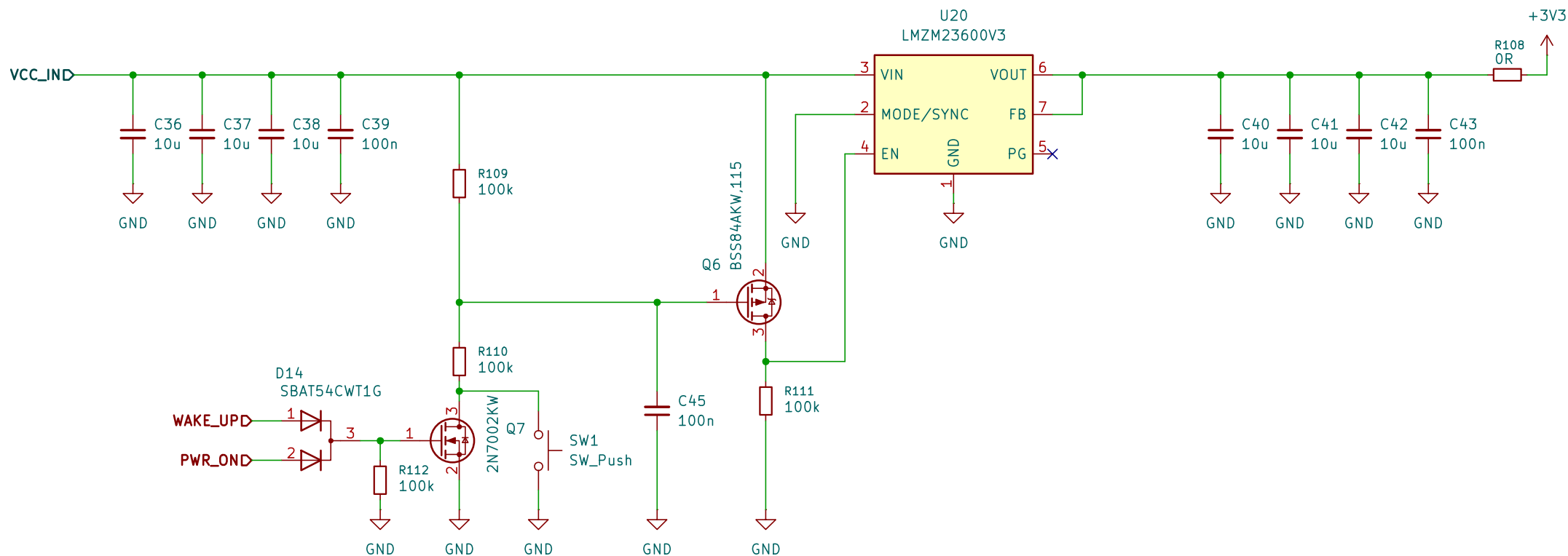
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Rev: B

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About:

- Buck converter generates 3.3V out of 4-35V, output up to 500mA.
- By default, power converter is off and whole system too. To enable it, signal from outside WAKE_UP needs to be set to 1. After that, once STM32 controller gets on, it can continue keeping power on by setting PWR_ON signal to 1.
- You can also enable power supply by using on-board switch



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Sheet: /Power/
File: Power.kicad_sch

Title: OpenBMS

Size: A4

Date: 2026-09-24

Rev: B

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Id: 7/7